

ALGEBRA TRICKS BY
EBENEZER
DOMPREH



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HOW TO SOLVE i^n (WHERE n IS A BIG NUMBER) WITHIN 7 SECONDS

Solution

Divide n by 4

Write your answer in the form A remainder B

Where A is the number of times 4 can equally divide n and B is the remainder

Your answer becomes i^B **NOTE**

$$i = \sqrt{-1}$$

Examples: i^{2003} **$i^2 = -1$**

$$i^{53} \text{ **$i^3 = -i$** }$$

$$2003/4 = 500r3$$

$$53/4 = 13r1$$

Ans: $i^3 = -i$

Ans: i



FINDING EIGEN VALUES WITHIN 30 SECONDS (FOR MCQS)

FACTS TO KNOW:

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- Product of the eigen values should be equal to the determinant of

the matrix **TIP**

EXAMPLE

- Change the calculator to

matrix mode

Find the eigen values of the matrix, $A = \begin{pmatrix} 4 & 1 \\ 1 & 4 \end{pmatrix}$
(**MODE + 6**)

A) 3, -3 C) -3, -5 • Press 1 for matrix A

B) 3, 5 D) 5, 0
matrix

□

- Key in the members of your

matrix

SOLUTION

- Press the on button

- Press **SHIFT + 4**

Trace of A = sum of leading diagonal members

- Press **7** for **det**

= 4 + 4 = 8 • Press **SHIFT + 4** again

- Select **MAT A** and press =

Determinant of A = $4(4) - 1(1) = 15$

To find the determinant of a matrix:

- Choose the dimension of your

***Visit **TIP** to know how to use the calculator to find the determinant of a*

matrix

SOLVING (A+Bi)ⁿ IN LESS THAN 1 MINUTE

Change your calculator to complex mode

To do this, press “MODE” then 2

- Type the complex number as you see it, on the calculator

To type i, press the **ENG** button. **NOTE**

Examples The highest power the normal calculator can solve with respect to

1. $(3+i)^5$
complex numbers is 3

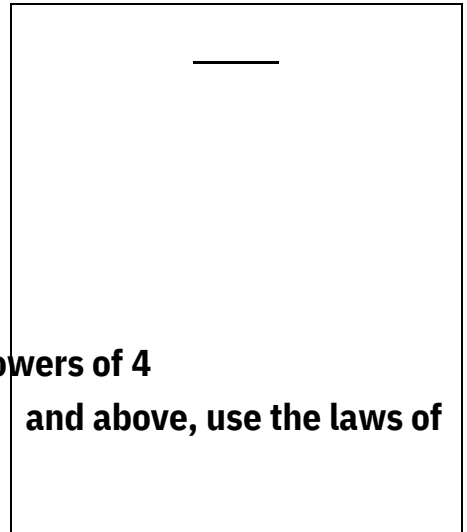
Type $(3+i)^3 * (3+i)^2$ on your calculator To solve for powers of 4

Because they have the same base, the powers add up
indices to split the

powers
2. $\frac{(4+3i)^5}{(3+i)^6}$

$$\frac{(\quad)}{(3+i)^3} * \frac{(\quad)}{(3+i)^3} \quad 4 + 3i \quad 4 + 3i \quad 2$$

Type on your calculator



FINDING THE NTH ROOT OF A COMPLEX NUMBER

- Convert the complex number (without the root) to the polar form
- Write your expression in the form
- $r^{\frac{1}{n}} [\cos(\frac{\Theta+2\pi k}{n}) + i \sin(\frac{\Theta+2\pi k}{n})]$

Where n is the root, k = 0 for the first root, k=1 for the second root, etc.

Example

$$\sqrt[2]{(8 + 4i)}$$

Solution

Convert the complex number (without the root) to polar form
 $r = 8.944$ $\Theta = 26.57$

Write your expression in the form

$$r^{\frac{1}{n}} [\cos(\frac{\Theta+2\pi k}{n}) + i \sin(\frac{\Theta+2\pi k}{n})]$$

1st root =

$$8.944^{\frac{1}{2}} [\cos(\frac{26.57+2\pi(0)}{2}) + i \sin(\frac{26.57+2\pi(0)}{2})] = 2.911 + 0.687i$$

2nd root =

$$8.944^{\frac{1}{2}} [\cos(\frac{26.57+2\pi(1)}{2}) + i \sin(\frac{26.57+2\pi(1)}{2})] = -2.911 - 0.687i$$

TIP

To convert a complex number to polar form using the calculator:

- Press **MODE** and select **2** for **COMPLX**
- Type your complex number in the rectangular form
- Press **SHIFT AND 2**
- Select option **3**

r represents the magnitude and

Θ is the angle

EXPRESSING POWERS OF TRIG FUNCTIONS IN SPECIFIED FORMS

WORKS FOR MCQS

EXAMPLE

Express $\cos^4 2x$ in terms of $\cos$ with an exponent of 1

- a) $\frac{1}{24} [3 + \cos^2 x + 4\cos 4x]$
- b) $\frac{1}{3} [3 + \cos^4 x + 2\cos 4x]$
- c) $[3 + \cos^2 x + 4\cos 4x]$
- d) $[3 + \cos^4 x + 4\cos 2x]$

SOLUTION

- Assume a value for x (advisable to choose 30, 45 or 60)
 - Put the value of the x chosen, one by one, in place of x in the question, and also in the options
 - The option that gives you the same results as the result of the question is the answer

ILLUSTRATION

Assume a value of x to be 30

$\cos^4 2x = \cos^4 2(30) = 0.0625$ For the question part

Please note that punching on the calculator, $\cos^4 2x = (\cos 2x)^4$

Try with $x = 30$ for the answer part. You will realise that only a give you the same result of 0.0625

$$\text{a) } \frac{1}{24} [3 + \cos^2 x + 4\cos 4x] = \frac{1}{24} [3 + \cos^2(30) + 4\cos 4(30)] = 0.0625$$

Therefore the answer is a

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